Fiscal Year:	FY 2012	**Task Last Updated:**	FY 07/11/2012
PI Name:	Rumschitzki, David Ph.D.		
Project Title:	How important is the contact angle in nucleate boiling heat transfer?		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date is now 6/30/2013, per NSSC information (Ed., 8/20/2012) NOTE: to be extended, per PI (Ed., 06/2012)		
Key Personnel Changes/Previous PI:	Continuing Postdoc: Lin Zheng, PhD. Continuing Undergraduate candidate: Mr. Omar Duran (minority student), 4th year BS candidate in Chemical Engineering.		
COI Name (Institution):	Lee, Taehun (City College of the City University of New York) Koplik, Joel (City College of the City University of New York)		
Grant/Contract No.:	NNX09AK40G		
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Task Description:	High latent heats allow evaporating small amounts of fluid to cool hot surfaces very efficiently. Boiling is thus attractive for cooling nuclear reactors, computer chips, hot surfaces in space (where volume is at a premium), etc,. provided the hot surface remains wet, which can be an issue in terrestrial and certainly is in microgravity. Dryout can be catastrophic. Despite decades of study, competing mechanisms for boiling heat transfer exist, including fluid mixing and liquid microlayer (under a portion of the vapor bubble) evaporation. We have carried out preliminary quasistatic heat conduction calculations of a single vapor bubble on the surface of a thin solid layer covered by a liquid of finite thickness. The solid sits on a heater at fixed temperature above the fluid's saturation temperature. The system is axisymmetric with periodic boundary conditions at its lateral edge to mimic bubble-bubble interactions. These conduction-only results match the pronounced heat transfer jump upon (surface) vapor bubble nucleation in all standard boiling curves and provide scalings of the heat flux with many key physical parameters that agree with historic correlations. Its simple mechanism may be transformative: The bubble's 3-phase contact line (CL) locally cools the superheated solid surface from an otherwise superheated value to the fluid saturation temperature. The contact line thus acts as a heat magnet from the solid for, e.g., microlayer vaporization. Critical questions include: How does the heat transfer vary as bubbles grow, deform, and detach? Does a detaching bubble leave behind a vapor residue/CL, generating a new bubble to draw heat? Does it detach cleanly? Or, does cooler fluid that replaces the rising bubble condense the residue bubble? A dynamic calculation addressing these questions needs CL motion with bubble growth as an input. Residue bubble quenching involves convection/convective transfer. We shall: 1. Create a molecular-sized model of our solid-liquid-vapor bubble system and let interparticle physics via molecular dynamics (MD) determine CL motion. We calculate bubble growth and heat transfer for a nanobubble directly from the MD results or input its CL motion into our quasistatic heat conduction model. This MD will begin with zero gravity and later introduce a pseudogravity so as to mimic a real bubble's surface tension-bouyancy balance (Bond number). 2. Formulate the fluid mechanics /heat transfer /phase change as a coarse-grained lattice-Boltzmann (LB) problem, which treats the liquid, its vapor and their mutual interface (including the CL) as a single fluid with a diffuse interface. One specifies the fluid-solid interaction energy; CL motion is part of the solution. This avoids the need to explicitly track the vapor-liquid interface and apply interfacial conditions there. In the presence of gravity, we use LB to follow deforming bubbles towards detachment for different system parameters, examine if a vapor residue remains on the surface after detachment and, if so, if convection quenches it. In gravity's absence we shall examine if the circulation provided by the expanding bubble's fluid displacement can help remove a bubble from the surface. We shall introduce intermittent fluid circulation from the bulk toward the surface along the domain boundary to push bubbles off and to convect them away from the surface. The goal is to avoid surface dryout, common in microgravity boiling, and to permit sustained, efficient surface cooling. If the calculations suggest this as being feasible, obvious microgravity/shuttle experiments would test it.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A vastly improved understanding of the mechanism of nucleate boiling heat transfer and the role played by gravity in it can have significant impacts on both terrestrial life and that on humans in space. An advance that makes nucleate boiling heat transfer essentially risk-free would enable the use of far less fluid to cool hot surfaces. For example, it could lead to the use of far less water in the secondary cooling of nuclear reactors, and therefore lead to less thermal pollution of our waters. Instead of releasing large amounts of hot water into rivers, one might be able to release far less hot water and more steam. Such an understanding might also influence computer design, since cooling issues are a critical constraint on computer design. In space applications, such cooling without the high risk of burnout could lead to far smaller cooling systems - and since the total space in the space station or in a space ship is extremely limited, such an advance could be highly beneficial.
	Taskbook Progress Report: for NASA grant #NNX09AK40G Title: How important is the contact line in nucleate boiling heat transfer? PI: David S. Rumschitzki, The City College of New York Period: 16 April 2011-15 April 2012 Extended Abstract: This project investigates the mechanism of heat transfer from a heater, through a thin solid to a liquid and a vapor bubble that are on the solid's surface. The solid conducts heat much better than the fluid. This study employs three methods/approaches to analyze this problem. The first solves the heat conduction equations, without buoyancy-driven fluid flow, for the instantaneous solid and liquid temperature and heat flux fields for a bubble in a liquid layer on the solid surface, with the bubble on the solid. The calculations show that, for high solid/liquid thermal conductivity, the overwhelming majority of the evaporation takes place at the three-phase contact line (CL) (unless the liquid above the bubble is also superheated). If one knows/ models the motion of the CL with bubble growth, a poorly-understood phenomenon even without evaporation at the CL, one can calculate how the bubble grows with time. Our preliminary work guessed two CL motion models and did this calculation for bubbles small enough to be sections of a sphere due to the predominance of interfacial tension forces over buoyancy at these scales. In year one, we extended this calculation to include the effects of gravity on slow bubble deformation and growth by allowing the bubble to grow to sizes where buoyancy is no longer negligible and where the bubble can assume non-spherical Young-Laplace shapes. These shapes balance surface tension and gravitational force when shear stresses are small. We then followed these bubbles to snap-off, appropriately defined. In order to avoid using only ad hoc models for the motion of the CL with the growth of the bubble's volume, we employed two methods in aims 2 and 3 that seek to find CL motion as an output of calculations that are based on more fundamental quantities that one might be able to specify far better than the CL motion itself. In year one we also begun to step down to the molecular level to so-called molecular dynamics (MD) calculations that model a nano-scale version of our solid-liquid-vapor system by prescribing the intermolecular interactions and solving the resulting Newton's equations of motion. The MD approach allows the system to tell us, as an output, what, among other things, the resulting CL motion is for various temperatures imposed at the lower face of the solid, temperatures above the fluid's saturation temperature. In contrast to the earlier continuum heat equation calculation, MD accounts for both conductive and convective heat transfer. In year one we succeeded in forming stable bubbles. In year two, we were able to grow this bubble by evaporation either with or without a uniform body force that worked in analogy to gravity (but was of far larger magnitude so as to be relevant at the nano-scale). For some reason the bubbles in year two grew to a stable size and did not grow further. In year two, we also began, in parallel, to use a lattice-Boltzmann (LB) method that treats both liquid and vapor as a single fluid of variable density to solve the flow (Navier Stokes) and energy equations with phase change and its associated thermodynamics. This approach also yields the CL position as an output of the calculation. Year

Task Progress:

two involved basic formulation of an LB approach to simultaneous fluid flow and heat transfer that included phase change and fluids (vapor/liquid) of quite different densities. Much of year two's LB work was focused on method development, since many of these features needed for this calculation, e.g., large density differences between phases, phase change, etc., went well beyond what had been done with LB in the past. We also concentrated on code development to incorporate these new features into the LB method in year two.

In year three we continued both the MD and LB approaches, whose results we summarize below. In our MD work, we solved the problem of stalled bubble growth in the presence of a uniform body force that appeared in year two. What had been happening in year two's studies was that evaporation at the CL was being balanced by condensation at the top of the bubble, where the temperature was lower. The temperature at the top wall of our system was set at the fluid's saturation temperature, but the imposition of a uniform body force meant that the local saturation temperature at points below the top wall was in fact higher than at the top wall. Adjusting the top wall temperature to the temperature at the top of the stalled bubble raised the temperature everywhere to values above the corresponding local saturation temperatures and allowed the bubble to grow, neck and detach. We followed these processes for different conditions and examined CL radius vs time, snapoff volume and, when it occurred, residue condensation. We also examined the heat flux and flow fields upon bubble growth and detachment, which give us detailed insights into the system's time evolution, including issues that impact whether and how the residue bubble condenses. In MD calculations, the issue of temperature slip, the violation of temperature continuity at the solid-fluid interface, a very small, but real, measurable effect in physical situations, had a pronounced effect on the problem. Whereas the liquid-vapor interface and, in particular, the CL was essentially at the saturation temperature in the liquid-vapor portion of the system– in agreement with the boundary conditions that we imposed in our continuum model, the temperature in the top layer of the solid at the CL was discontinuously higher and not always significantly cooler than its surrounding, otherwise-superheated solid. That is, it did not always correlate directly with the temperature distribution in the adjacent lowest fluid layer. This non-continuity clearly influences the correspondence between the continuum and MD models, where the former assumes temperature continuity, i.e., no temperature slip, and finds the cooled (to the saturation temperature) CL causes the CL to be a very strong heat magnet/sink for heat flow in the otherwise superheated solid. Despite this drawback, many interesting features of the simulation, e.g., CL radius vs time leading to snapoff, agree well with experiment.

Our LB work has progressed significantly. We first verified the code by comparing its results with those of problems that have known exact similarity solutions, e.g., Stephan's problem (a one-dimensional moving front of phase change) and Scriven's problem (a spherical vapor bubble growing by evaporation in an unbounded liquid), and the agreement is excellent. We then used our code to study the (two-dimensional) unsteady growth of a vapor bubble in a liquid layer on an infinitely thin solid, followed by the analogous growth on a solid of finite thickness, both heated at constant temperature from below. The difference between the two cases is that, in the former, the uniform temperature condition is applied to the solid surface in contact with the vapor, liquid and, in particular, the CL, thereby, by hypothesis, contradicting the assumptions of the continuum model that the CL is at the saturation temperature. In the continuum model, we apply this uniform temperature condition on the lower face of the solid, and therefore the temperature distribution at the solid-fluid interface is an output of the calculation and satisfies continuity there. In each of these calculations we begin with a small bubble that was a section of a sphere and follow it to detachment for various values of the non-dimensional quantities, the Morton and Bond numbers, as well as a gradient parameter that controls both the surface tension and the liquid-vapor interface thickness. In this diffuse interface method, the liquid and vapor are treated as a single fluid of variable density at the cost of the interface having a bounded density gradient, i.e., an interface of non-infinitesimal thickness. As a result, raising the surface tension thickens the interface. Since the assumption that the interface maintains the saturation temperature is a local equilibrium argument, and such rapid equilibrium depends on the interfacial length scale being very small, the thick interfaces, despite achieving the saturation temperature far from the solid surface, indeed deviates from it near to it, where the heat flux is large. Nevertheless, the CL is still at the minimum temperature along the solid-fluid interface, and therefore still acts as a strong heat sink through the solid, as was the case in the continuum model. In the parameter range where the calculations converge (Morton number 10-4 or higher; note the Morton number for water ~10-10; Bond numbers ~.02-0.2) thus far, the residue bubble is usually comparable or only slightly smaller in size with the escaped bubble size, unlike the experiments that take place in a very different parameter regime. Study of the snapoff process at these parameters yields insights into the residue bubble, the flow and heat flux fields. We are currently focusing on understanding the situations where the residue bubble either condenses or grows to form a new bubble and on the conditions that transform the system from one that yields individual bubbles to one where the solid surface is covered by a vapor layer. These transitions may be provide insight into the experimentally-observed transitions from single bubble to jets and columns subregions of nucleate boiling and from nucleate to film boiling.

Invited talks:

"Vapor bubble growth, deformation & detachment from a heating surface: The contact line and the conjugate problem," NASA Glenn Research Center, Cleveland, Ohio, March 2012.

Talks at conferences:

Bao, J., Koplik, J. and Rumschitzki, D.S., "Simulation of a vapor bubble growth in nucleate boiling," 2009 AIChE Annual Meeting, Nashville, Tenn, November 2009.

Bao, J., Koplik, J. and Rumschitzki, D.S., "Molecular dynamics of heterogeneous bubble nucleation and growth: Study of contact line motion relevant to nucleate pool boiling," 2010 AIChE Annual Meeting, Salt Lake City, Utah, November 2010.

Bao, J., Huang, L. Dagan, Z. and Rumschitzki, D., "Heat conduction to a vapor bubble in a liquid layer on a thin solid heated from below: The contact line, the conjugate problem. Continuum and molecular dynamics results," 49th European Two-Phase Flow Group Meeting, Tel Aviv, Israel, 26 May-2 June 2011.

Zheng, L. , Lee, T., Guo, Z. and Rumschitzki, D. , "Parasitic Shrinkage of Bubbles/Drops in the Lattice Boltzmann Equation Method for Nonideal Gases," The 20th Discrete Simulation of Fluid Dynamics (DSFD), Fargo, ND, Aug. 8–12 (2011).

Bao, J., Koplik, J. and Rumschitzki, D.S., "Molecular dynamics of heterogeneous vapor bubble growth: Exploring the physics of the 3-Phase contact line," 2011 AIChE Annual Meeting, Minneapolis, MN, 16-21 October 2012.

Rumschitzki, D., "How the microphysics of three-phase contact line motion is important in nucleate boiling," The 6th Sino-US Joint Conference of Chemical Engineering, Beijing, China, 7-10 November 2011.

Papers to be submitted for journal publication:

1 Bao as lead author: On the continuum model for a bubble in a liquid layer on a solid that is heated from below: static results and comparison with historic correlations on heat flux

	2. Bao as lead author: On the continuum model for the growth of a bubble in a liquid layer on a solid that is heated from below: Bubble growth and comparison with experiment. 3. Bao as lead author: Molecular dynamics study of the growth of a vapor bubble in a liquid layer on a solid that is heated from below: What we can learn about contact line motion and bubble growth and detachment. 4. Zheng as lead author: A lattice Boltzmann study of the growth of a vapor bubble in a liquid layer on an infinitely thin superheated solid surface. 5. Zheng as lead author: A lattice Boltzmann study of the growth of a vapor bubble in a liquid layer on a solid heated from below: Bubble growth and detachment and its relevance to the transition between the single bubble and jets and columns sub-regions of nucleate boiling. .
Bibliography Type:	Description: (Last Updated: 04/09/2020)
Abstracts for Journals and Proceedings	Bao J, Koplik J, Rumschitzki DS. "Simulation of a vapor bubble growth in nucleate boiling." Nucleation and Growth session. Presented at the 2009 AIChE Annual Meeting, Nashville, Tennessee, November 8-13, 2009. 2009 AIChE Annual Meeting, Nashville, Tennessee, November 8-13, 2009. http://www3.aiche.org/Proceedings/Abstract.aspx?ConfID=Annual-2009&GroupID=1356&SessionID=10160&PaperID=165442 , Nov-2009
Abstracts for Journals and Proceedings	Bao J, Koplik J, Rumschitzki DS. "Molecular dynamics of heterogeneous bubble nucleation and growth: Study of contact line motion relevant to nucleate pool boiling." Nucleation and Growth session. 2010 AIChE Annual Meeting, Salt Lake City, Utah, November 7-12, 2010. 2010 AIChE Annual Meeting, Salt Lake City, Utah, November 7-12, 2010. http://www3.aiche.org/Proceedings/Abstract.aspx?ConfID=Annual-2010&GroupID=1503&SessionID=14014&PaperID=199758 , Nov-2010
Abstracts for Journals and Proceedings	Bao J, Huang L, Dagan Z, Rumschitzki D. "Heat conduction to a vapor bubble in a liquid layer on a thin solid heated from below: The contact line, the conjugate problem. Continuum and molecular dynamics results." Presented at the 49th European Two-Phase Flow Group Meeting, Tel Aviv, Israel, May 29-June 2, 2011. 49th European Two-Phase Flow Group Meeting, Tel Aviv, Israel, May 29-June 2, 2011. , Jun-2011
Abstracts for Journals and Proceedings	Bao J, Koplik J, Rumschitzki DS. "Molecular dynamics of heterogeneous vapor bubble growth: Exploring the physics of the 3-Phase contact line." Nucleation and Growth session. Presented at the 2011 AIChE Annual Meeting, Minneapolis, MN, October 16-21, 2011. 2011 AIChE Annual Meeting, Minneapolis, MN, October 16-21, 2011. http://www3.aiche.org/Proceedings/Abstract.aspx?ConfID=Annual-2011&GroupID=1553&SessionID=17187&PaperID=238139 , Oct-2011
Abstracts for Journals and Proceedings	Zheng L, Lee T, Guo Z, Rumschitzki D. "Parasitic Shrinkage of Bubbles/Drops in the Lattice Boltzmann Equation Method for Nonideal Gases." 20th Discrete Simulation of Fluid Dynamics. DSFD 2011, Fargo, ND, August 8-12, 2011. The 20th Discrete Simulation of Fluid Dynamics. DSFD 2011, Fargo, ND, August 8-12, 2011. , Aug-2011
Dissertations and Theses	Bao J (Jinyong Bao, PhD). "The motion of the three-phase contact line and its role in heat transfer in the growth of a vapor bubble on a heated solid surface." PhD. Dissertation, Department of Chemical Engineering, The City University of New York, August 2011. , Aug-2011